

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 - SHORT ANSWER TEST

Course Code:	CSA0305	Course Title:	Data Structures
CO Assessed:	CO5 - Manipulation (BL3, BL4)	Assessment:	Assessment Tool 3 - Short Answer Test
Weightage:	30% of CIA	Total Marks:	100
CO2 Statement:	Construct MST using shortest path algorithms and traverse the graph to model and analyse real-world problem.		
Student Name:	K. Kabilan	Reg. No.:	192521157
Section / Batch:	B Tech IT	Date:	04/09/2026

Code of Conduct

I K. Kabilan (192521157) (Name / Reg No)

certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: K. Kabilan

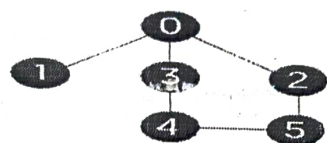
Instructions

- This test contains 10 short answer test questions. Total: 100 Marks.
- When a student answer using an Analytical problem-solving, in evaluation the answer should be with following five requirements: Understanding of problem, select appropriate data structure, Design the solution, analyze, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.
- No DTP printouts or electronic devices permitted. They should provide written materials.

Section / Topic	Focus	Q Nos	Marks
Minimum Spanning Tree	Kruskal's Algorithm	1	20
Graph Sorting	Topological Sort	2	20
Shortest Path Algorithm	MST & Dijkstra's Algorithm	3	20
Shortest Path Algorithm	Dijkstra's Algorithm	4	20
Minimum Spanning Tree	Prim's or Kruskal's Algorithm	5	20
GRAND TOTAL:			100 Marks

Annexure A - Short Answer Test

- Perform the following operations using stack. Assume the size of the stack is 5 and have a value of 22, 55, 33, 66, 88 in the stack from 0 position to size-1. Now perform the following operations: 1) Invert the elements in the stack, 2) Pop (), 3) Pop (), 3) Pop Construct AVL tree for the following elements. Mention the
- Given a collection of elements 10,20,30,40,50,60,70,80. The Key element to be search 70. Compare how searching is performed in Linear Search and Binary Search. Identify the efficient search method. Justify.
- What is the postfix and prefix forms of the following expression? $A+B*(C-D)/(P-R)$
- Construct AVL tree for the following elements. Mention the Rotations in each step of insertion. 1,2,3,4,5,6,7,8,9,10
- Construct a heap by inserting 10, 15, 12, 23, 21, 1, 6, 43, 34, 56, 78, 23, 45 into a binary heap.
- Construct a B-Tree of Order 3 by inserting numbers 34,26,45,12,87,56,78,22,23,24,65,85,95.
- Perform the merge sort for the following inputs. Also, show the result for each step of iteration. 40,20,70,14,60,61,97,30.
- How the following data set can be sorted by quick sort. Show each step-in sorting. 78,23,31,88,43,55,67,55.
- Perform Open Hashing & Linear Probing for the given set of 0,1,81, 4,64,25,16,36, 9 and 49.
- Consider the following graph & construct BFS.



Graph with six nodes

Rubrics for Calculation

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Understanding	30	Demonstrates complete and accurate understanding of concepts	Good understanding with <u>minor gaps</u>	Basic understanding	Limited understanding
Accuracy of Answer	25	Answers are completely correct and relevant	Mostly correct with <u>minor errors</u>	Partially correct	Mostly incorrect
Problem-Solving & Reasoning	20	Provides logical reasoning and appropriate steps	Good reasoning with <u>minor gaps</u>	Basic reasoning	Lacks logical reasoning
Clarity & Relevance	15	Answers are precise, clear, concise, and directly address the question	Mostly clear and relevant	Somewhat unclear or incomplete	Irrelevant or unclear answers
Time Management & Completion	10	Completes all questions accurately within the allotted time	Completes <u>most</u> questions on time	Requires additional time	Unable to complete the test

Faculty In-charge	Course Coordinator	Program Director
Signature: <u>[Signature]</u>	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Stack operation

Given stack : 22, 55, 33, 66, 88

Assume 88 is the Top before inversion

Top $\rightarrow$ 88
66
33
55
22

operation: 1 Invert the stack
After reversing all element

Top $\rightarrow$ 22
55
33
66
88

operation 2: pop()

Popped = 22

Stack: Top $\rightarrow$ 55
33
66
88

operation: 3 pop()

Popped = 55

Stack: Top $\rightarrow$ 33
66
88

operation: 4: pop()

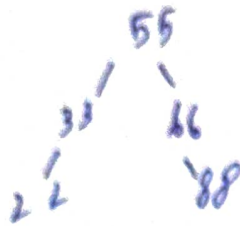
Popped = 33

Final stack: Top $\rightarrow$ 66
88

Final Stack = 91, 66

ii) AVL Tree

22, 55, 93, 66, 88



2) Linear Search

Elements: 10, 20, 30, 40, 50, 60, 70, 80

Search key: 70

Compare one by one

10 → 20 → 30 → 40 → 50 → 60 → 70

Comparisons = 7

Binary Search

Middle = 40 → 70 > 40

Search high 50, 60, 70, 80

Middle = 60 → 70 > 60

right half = 70, 80

Middle = 70 → found

Comparison = 3.

∴ Binary Search is more efficient because it reduces the search space by half at each step.

Linear search: $O(n)$

Binary Search: $O(\log n)$

Therefore Binary search is the efficient method for sorted data.

infix expression
 $A + B \times (C - D) / (P - R)$

Postfix

$ABCD - * PR - / +$

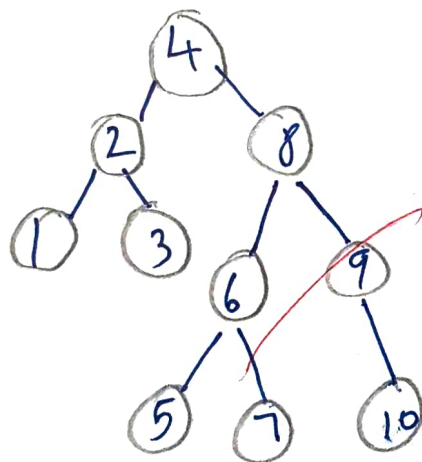
$= ABCD - * PR - / +$

Prefix

$+ A / * B - C D - P R$

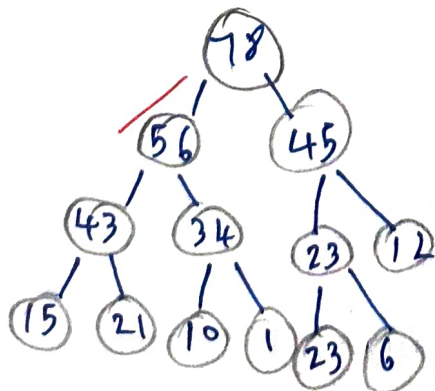
infix $+ / B - C D - P R$

4) Final AVL Tree



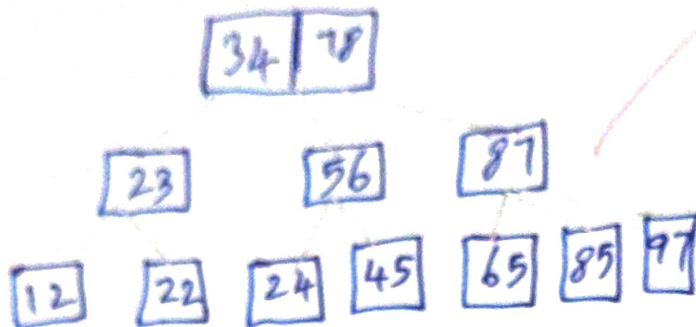
5. Max Heap (largest element at the root)

10, 15, 12, 23, 21, 1, 6, 43, 34, 56, 78, 23, 45



6. B-Tree of order 3
 maximum children = 3
 maximum keys = 2

34, 26, 45, 12, 87, 56, 78, 22, 23, 24, 65, 85, 95



Merge Sort

[40, 20, 70, 14, 60, 61, 97, 30]

↓

[40, 20, 70, 14]

[60, 61, 97, 30]

↓

[40, 20] [70, 14]

[60, 61] [97, 30]

40 20 70 14

60 61 97 30

Merge

[40] + [20] → [20, 40]

[70] + [14] → [14, 70]

[60] + [61] → [60, 61]

[97] + [30] → [30, 97]

Merge

[20, 40] + [14, 70] → [14, 20, 40, 70]

[60, 61] + [30, 97] → [30, 60, 61, 97]

²
 $[14, 20, 40, 70] + [30, 60, 61, 97] \rightarrow [14, 20, 30, 40, 60, 61, 70, 97]$

¹
Pick Sort

Given 78, 23, 31, 88, 43, 55, 67, 55

Pivot = 55

Sort left and right part

23, 31, 43 left sort

Right sort

88, 78, 67, 55

Pivot - 67

~~67, 78, 88~~

Final Sorted array

23, 31, 43, 55, 55, 67, 78, 88

9.

Element	Hash	position
0	0	0
1	1	1
81	1	2 (collision)
4	4	4
4	4	5 (collision)
64	4	6 (collision)
25	5	7 (collision)
16	6	8 (collision)
36	6	
9	9	9
49	9	3 (collision)

10) Assuming BFS starts from node 0:

Step	Queue	visited
1	0	0
2	1, 3, 2	0, 1, 3, 2
3	3, 2	0, 1, 3, 2
4	2, 4	0, 1, 3, 2, 4
5	4, 5	0, 1, 3, 2, 4, 5

BFS Traversal

~~0~~ → 1 → 3 → 2 → 4 → 5